

SKILL WEEK5

TASK 1 — Single Quotes

To apply single quotes, preserve literal content, ignore variable expansion, store quoted strings, validate parsing results, and test edge cases.

```
#!/bin/bash

echo "===== TASK 1: SINGLE QUOTES ====="

name="Laasya"
message='Hello $name'

echo "Variable name: $name"
echo "Message using single quotes: $message"

echo
echo "Testing literal content:"
text='This is a $variable and it is not expanded.'
echo "$text"

echo
echo "Testing spaces:"
sentence='Ubuntu shell scripting preserves all spaces inside single quotes.'
echo "$sentence"

echo
echo "Testing special characters:"
special='Special characters: $HOME * ? [ ] { }'
echo "$special"

echo
echo "Testing quoted string storage:"
quoted='This entire sentence is stored as one string.'
echo "Stored string: $quoted"

echo
echo "===== PARSING VALIDATION ====="

if [ "$message" = 'Hello $name' ]; then
    echo "PASS: Variable was not expanded inside single quotes."
else
    echo "FAIL: Variable was expanded."
```

```

echo
echo "===== PARSING VALIDATION ====="

if [ "$message" = 'Hello $name' ]; then
    echo "PASS: Variable was not expanded inside single quotes."
else
    echo "FAIL: Variable was expanded."
fi

if [ "$special" = 'Special characters: $HOME * ? [ ] { }' ]; then
    echo "PASS: Special characters preserved literally."
else
    echo "FAIL: Special characters changed."
fi

echo
echo "===== EDGE CASE TESTS ====="

empty=''
echo "Empty string length: ${#empty}"

spaces='    Text with spaces    '
echo "String with spaces: [$spaces]"

echo
echo "Task 1 completed successfully."

```

```

laasya@laasya:~$ cd skillweek5
laasya@laasya:~/skillweek5$ nano task1_single_quotes.sh
laasya@laasya:~/skillweek5$ chmod +x task1_single_quotes.sh
laasya@laasya:~/skillweek5$ ./task1_single_quotes.sh
===== TASK 1: SINGLE QUOTES =====
Variable name: Laasya
Message using single quotes: Hello $name

Testing literal content:
This is a $variable and it is not expanded.

Testing spaces:
Ubuntu shell scripting preserves all spaces inside single quotes.

Testing special characters:
Special characters: $HOME * ? [ ] { }

Testing quoted string storage:
Stored string: This entire sentence is stored as one string.

===== PARSING VALIDATION =====
PASS: Variable was not expanded inside single quotes.
PASS: Special characters preserved literally.

===== EDGE CASE TESTS =====
Empty string length: 0
String with spaces: [    Text with spaces    ]

Task 1 completed successfully.
laasya@laasya:~/skillweek5$ |

```

TASK 2 — Double Quotes

To apply double quotes, preserve spaces, allow variable expansion, parse nested tokens, validate outputs, and test quoted commands.

```
#!/bin/bash|

echo "===== TASK 2: DOUBLE QUOTES ====="

name="Laasya"
course="Computer Science"
year=2

echo "Basic variable expansion:"
echo "Name: $name"
echo "Course: $course"
echo "Year: $year"

echo
echo "===== PRESERVING SPACES ====="

message="Hello $name, welcome to Ubuntu shell scripting."
echo "$message"

sentence="This is a sentence containing multiple words."
echo "Sentence: $sentence"

echo
echo "===== VARIABLE EXPANSION ====="

college="BTech"
student="$name"
details="$student is studying $college."

echo "$details"

echo
echo "===== NESTED TOKENS ====="

language="Bash"
topic="Double Quotes"
```

```
info="$language is being used to study $topic."
echo "$info"

echo
echo "===== QUOTED COMMAND ====="

file="sample file.txt"

echo "Creating a file with spaces in its name..."
touch "$file"

if [ -f "$file" ]; then
    echo "PASS: File created successfully."
else
    echo "FAIL: File was not created."
fi

echo "Removing the test file..."
rm "$file"

echo
echo "===== OUTPUT VALIDATION ====="

expected="Hello $name, welcome to Ubuntu shell scripting."

if [ "$message" = "$expected" ]; then
    echo "PASS: Double quotes preserved the complete string."
else
    echo "FAIL: Output does not match expected value."
fi

echo
echo "===== EDGE CASE TESTS ====="

empty=""
echo "Empty string: [$empty]"
```

```
laasya@laasya:~/skillweek5$ nano task2_double_quotes.sh
laasya@laasya:~/skillweek5$ chmod +x task2_double_quotes.sh
laasya@laasya:~/skillweek5$ ./task2_double_quotes.sh
===== TASK 2: DOUBLE QUOTES =====
Basic variable expansion:
Name: Laasya
Course: Computer Science
Year: 2

===== PRESERVING SPACES =====
Hello Laasya, welcome to Ubuntu shell scripting.
Sentence: This is a sentence containing multiple words.

===== VARIABLE EXPANSION =====
Laasya is studying BTech.

===== NESTED TOKENS =====
Bash is being used to study Double Quotes.

===== QUOTED COMMAND =====
Creating a file with spaces in its name...
PASS: File created successfully.
Removing the test file...

===== OUTPUT VALIDATION =====
PASS: Double quotes preserved the complete string.

===== EDGE CASE TESTS =====
Empty string: []
Number stored as string: [12345]
Special characters: Characters: $HOME * ?

Task 2 completed successfully.
```